



Cambridge IGCSE™

PHYSICS

0625/42

Paper 4 Extended Theory

May/June 2023

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

Cambridge International is publishing the mark schemes for the May/June 2023 series for most Cambridge IGCSE, Cambridge International A and AS Level and Cambridge Pre-U components, and some Cambridge O Level components.

PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

Science-Specific Marking Principles

1	Examiners should consider the context and scientific use of any keywords when awarding marks. Although keywords may be present, marks should not be awarded if the keywords are used incorrectly.
2	The examiner should not choose between contradictory statements given in the same question part, and credit should not be awarded for any correct statement that is contradicted within the same question part. Wrong science that is irrelevant to the question should be ignored.
3	Although spellings do not have to be correct, spellings of syllabus terms must allow for clear and unambiguous separation from other syllabus terms with which they may be confused (e.g. ethane / ethene, glucagon / glycogen, refraction / reflection).
4	The error carried forward (ecf) principle should be applied, where appropriate. If an incorrect answer is subsequently used in a scientifically correct way, the candidate should be awarded these subsequent marking points. Further guidance will be included in the mark scheme where necessary and any exceptions to this general principle will be noted.
5	<u>'List rule' guidance</u> For questions that require <i>n</i> responses (e.g. State two reasons ...): • The response should be read as continuous prose, even when numbered answer spaces are provided. • Any response marked <i>ignore</i> in the mark scheme should not count towards <i>n</i>. • Incorrect responses should not be awarded credit but will still count towards <i>n</i>. • Read the entire response to check for any responses that contradict those that would otherwise be credited. Credit should not be awarded for any responses that are contradicted within the rest of the response. Where two responses contradict one another, this should be treated as a single incorrect response. • Non-contradictory responses after the first <i>n</i> responses may be ignored even if they include incorrect science.

6 Calculation specific guidance

Correct answers to calculations should be given full credit even if there is no working or incorrect working, **unless** the question states 'show your working'.

For questions in which the number of significant figures required is not stated, credit should be awarded for correct answers when rounded by the examiner to the number of significant figures given in the mark scheme. This may not apply to measured values.

For answers given in standard form (e.g. $a \times 10^n$) in which the convention of restricting the value of the coefficient (a) to a value between 1 and 10 is not followed, credit may still be awarded if the answer can be converted to the answer given in the mark scheme.

Unless a separate mark is given for a unit, a missing or incorrect unit will normally mean that the final calculation mark is not awarded. Exceptions to this general principle will be noted in the mark scheme.

7 Guidance for chemical equations

Multiples / fractions of coefficients used in chemical equations are acceptable unless stated otherwise in the mark scheme.

State symbols given in an equation should be ignored unless asked for in the question or stated otherwise in the mark scheme.

Acronyms and shorthand in the mark scheme

Acronym / shorthand	Explanation
A mark	Final answer mark which is awarded for fully correct final answers including the unit.
C mark	Compensatory mark which may be scored when the final answer (A) mark for a question has not been awarded.
B mark	Independent mark which does not depend on any other mark.
M mark	Method mark which must be scored before any subsequent final answer (A) mark can be scored.
Brackets ()	Words not explicitly needed in an answer, however if a contradictory word / phrase / unit to that in the brackets is seen the mark is not awarded.
<u>Underlining</u>	The underlined word (or a synonym) must be present for the mark to be scored. If the word is a technical scientific term, the word must be there.
/ or OR	Alternative answers any one of which gains the credit for that mark.
owtte	Or words to that effect.
ignore	Indicates either an incorrect or irrelevant point which may be disregarded, i.e., <u>not</u> treated as contradictory.
insufficient	An answer not worthy of credit <u>on its own</u> .
CON	An incorrect point which contradicts any correct point and means the mark cannot be scored.
ecf [question part]	Indicates that a candidate using an erroneous value from the stated question part must be given credit here if the erroneous value is used correctly here.
cao	Correct answer only.
ORA	Or reverse argument.

Question	Answer	Marks
1(a)(i)	no resultant / net force	B1
	no resultant/net moment	B1
1(a)(ii)	$4.7 \times 10^7 \text{ J}$ or 47 MJ	A2
	$(\Delta)E_p = mg(\Delta)h$ OR $(\Delta E_p =) mg(\Delta)h$ OR $(\Delta E_p =) 3200 \times 9.8 \times 1500$	C1
1(b)(i)	point, labelled 1, on either of the horizontal sections of the graph (to the left of A or to the left of B)	B1
	point, labelled 2, on the graph between A and the start of the horizontal section of the graph to the left of B	B1
	point, labelled 3, on the graph between the start of the curved section to the right of the origin and the start of the horizontal section of the graph to the left of A	B1
1(b)(ii)	(initially there is acceleration due to) weight OR gravitational force OR unbalanced force / resultant force / downward force	B1
	(then) air resistance increases as speed or velocity increases	B1
	(as air resistance increases) resultant force downwards decreases OR acceleration decreases	B1
	constant speed when air resistance = weight / gravitational force	B1

Question	Answer	Marks
2(a)	26 J	A3
	$E_K = \frac{1}{2}mv^2$ OR $(E_K =) \frac{1}{2}mv^2$ OR $(E_K =) \frac{1}{2} \times 0.16 \times (18)^2$	C1
	$(E_K =) \frac{1}{2} \times 0.16 \times (18)^2$ OR $(E_K =) \frac{1}{2} \times 0.16 \times 324$ OR $(E_K =) 2.6 \times 10^N$	C1

Question	Answer	Marks
2(b)	24 N	A2
	$Ft = \Delta mv$ OR $F = ma$ OR $(F =) (0.16 \times 18) / 0.12$	C1
2(c)	longer time (of impact/contact) AND smaller force (on them) OR longer time (of impact/contact) AND does not hurt as much	B1

Question	Answer	Marks
3(a)	(force of gravity / weight of person is spread over a much) greater <u>area</u>	B1
	$p = F/A$ OR $p \propto 1/A$	B1
	(force is same so) pressure is lower (so ice is less likely to crack)	B1
3(b)	$5.8 \times 10^3 \text{ Pa}$	A4
	p (due to water) = ρgh OR $(p =) \rho gh$ OR $(p =) 1000 \times 9.8 \times 0.45$ OR $(p =) 4410$	C1
	$W = mg$ OR $(W =) mg$ OR $(W =) (690 \times 9.8)$ OR $(W =) 6762$ OR $(p$ (due to ice) =) 1352.4	C1
	(pressure =) candidate's calculated pressure due to water + candidate's calculated pressure due to ice OR total pressure = $[1000 \times 9.8 \times 0.45] + [(690 \times 9.8) / 5.0]$ OR total pressure = 4410 + 1352.4	C1

Question	Answer	Marks
4(a)	pressure decreases AND particles have smaller velocity / momentum / smaller E_k / kinetic energy (when temperature is lower)	B1
	lower rate / frequency of collision of particles	B1
	particles collide with smaller force OR smaller impulse change	B1

Question	Answer	Marks
4(b)(i)	$-273\ (^{\circ}\text{C})$	B1
4(b)(ii)	(temperature at which) particles have least E_k / kinetic energy	B1
	lowest possible temperature	B1
4(c)	200 cm^3	A3
	$pV = \text{constant}$ OR $9.0 \times 10^4 \times 350 = 1.6 \times 10^5 \times V_2$	C1
	$V_2 = [9.0 \times 10^4 \times 350] / 1.6 \times 10^5$ OR $V_2 = 2.0 \times 10^N$ OR 1.97×10^N	C1

Question	Answer	Marks
5(a)(i)	$\rho = m / V$ OR $m = \rho V$	B1
	$(m =) 1.2 \times 4.5 \times 6.1 \times 2.4 (= 79\text{ kg})$ OR $(m =) 79.056\text{ (kg)}$	B1
5(a)(ii)	290 s	A4
	$c = (\Delta)E / m\Delta\theta$ OR $(\Delta E =) mc\Delta\theta$ OR $(\Delta E =) 79 \times 1000 \times 4(.0)$ OR $(\Delta E =) 316\ 000$ OR $(\Delta\theta =) 4(.0)$	C1
	$P = (\Delta)E / t$ OR $(\Delta E =) Pt$ OR $(\Delta E =) 1100 \times t$	C1
	$(t =) mc\Delta\theta / P$ OR $(t =) 79 \times 1000 \times 4(.0) / 1100$ OR $(t =) 316\ 000 / 1100$	C1
5(a)(iii)	any one from: (thermal) energy is transferred to furniture / walls / objects (in the room) (thermal) energy is transferred through windows / doors / floor / ceiling / from the room	B1
5(b)	conduction AND convection	B1

Question	Answer	Marks
6(a)	P-waves: longitudinal	B1
	S-waves: transverse	B1
6(b)	1600 m OR 1.6 km	A3
	$v = f\lambda$ OR ($\lambda =$) v/f	C1
	($\lambda =$) $[7.2 \times 1000] / 4.5$ OR ($\lambda =$) 1.6×10^N OR ($\lambda =$) $7.2 / 4.5$	C1

Question	Answer	Marks
7(a)	normal drawn in correct position and at right angles to the surface	B1
7(b)	22°	A3
	$i = 34(^\circ)$	C1
	$n = \sin i / \sin r$ OR ($r =$) $\sin^{-1} \{\sin i / n\}$ OR $\sin r = \sin 34 / 1.47$ OR $\sin r = 0.38$	C1
7(c)	$3.0 \times 10^8 \text{ m/s}$	B1
7(d)	$2.04 \times 10^8 \text{ m/s}$	A2
	$n = \text{speed of light in air} / \text{speed of light in oil}$ OR ($\text{speed of light in oil} =$) $\text{speed of light in air} / n$ OR ($\text{speed of light in oil} =$) $3.0 \times 10^8 / 1.47$	C1

Question	Answer	Marks
8(a)(i)	region in which a (magnetic) pole experiences a force	B1
8(a)(ii)	in the direction of the force on the N pole	B1

Question	Answer	Marks
8(b)	4 radial lines outside sphere, touching the sphere and equally spaced all around sphere	B1
	direction of arrows towards the sphere	B1
8(c)(i)	2.0 V	A1
8(c)(ii)	(ratio of p.d. across $R_2 : R_3 =$) 1 : 2	B1
8(c)(iii)1.	current is zero in R_1 AND diode is in wrong direction (to allow current) owtte	B1
8(c)(iii)2.	(ratio of p.d. across $R_2 : R_3 =$) 1 : 1	B1

Question	Answer	Marks
9(a)	α – no. of neutrons 2	B1
	β – no. of protons 0 and charge -1.6×10^{-19}	B1
	γ – no. of neutrons 0 and charge 0 and (very) thick concrete / thick lead	B1
9(b)	(the nucleus has) one less neutron and one more proton	B1
9(c)	95 (counts / min)	A4
	initial count rate due to source = $550 - 30$ (counts / min) OR 520 seen	C1
	(75 min =) 3 half-lives OR (count rate =) $1/8$ (of initial count rate)	C1
	final count rate due to source = $(520 / 8 =)$ 65	C1
9(d)	any two from: • limit time of exposure • store sources in lead boxes • keep distance from sources • avoid contact OR use tongs OR wear gloves	B2

Question	Answer	Marks
10(a)	$v = 2\pi r / T$	B1
	$r =$ (average) radius of the <u>orbit</u> AND $T =$ (orbital) period	B1
10(b)	rays from Sun strike the country at different angles through the year OR rays from Sun strike the country for different number of hours per day through the year	B1
10(c)	(first space:) red supergiant	B1
	(second space:) nebula	B1
	(3 rd and 4 th spaces:) neutron star	B1
	black hole	B1
10(d)	1.6×10^9 (light-years)	A2
	$H_0 = v/d$ OR $(d =) v / H_0$ OR $(d =) [33\,000 \times 10^3] / [2.2 \times 10^{-18} \times 9.5 \times 10^{15}]$	C1